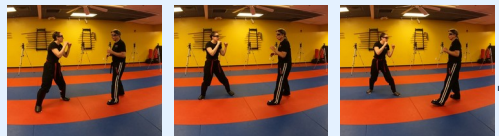


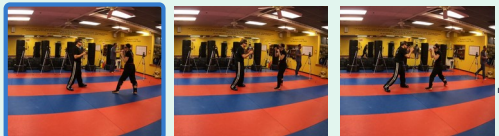
Multi-shot video

Shot 1



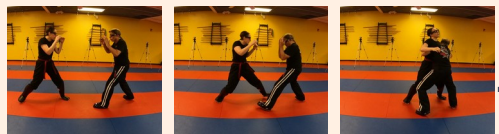
Shot transition

Shot 2



Shot transition

Shot 3



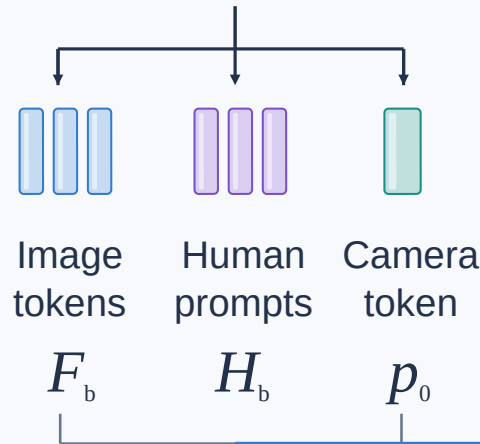
Current + past RGB
Online shot-transition detector

Feature & token construction

New-shot frame I_b



Image encoder



Same boundary-frame tokens

Shot-transition signal

❄️ Frozen 🔥 Trainable

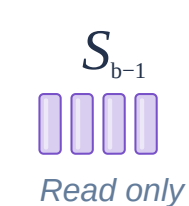
Alignment-state decoupling

Repeated at each detected transition

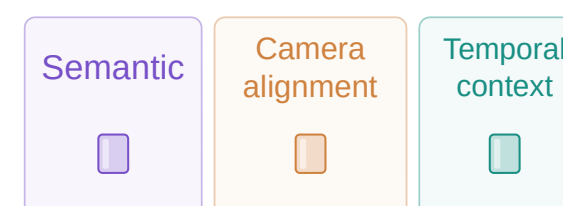
Alignment path

History-conditioned · Temporary

Previous state S_{b-1}



Alignment token construction



Native tokens

Z_b^{align}

Recurrent decoder

Camera latent residual

$\tilde{C}_b$

$\tilde{S}_b$ - X Not propagated

Reconstruction path

Reset at cut · Updated within shot

Initial state $S_\emptyset$



Native tokens

Recurrent decoder

Camera C_b^r

Humans $\mathcal{H}_b^r$

State S_b^r

New-shot outputs

Later-frame tokens $t > b$

Recurrent state update for $t > b$

Camera-human consistency

Camera alignment

$$T_b^{\text{align}} = \tilde{C}_b (C_b^r)^{-1}$$



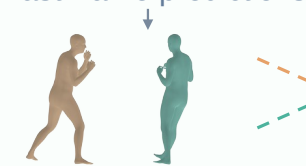
T_b^{align}



Geometry-based person association

Previous shot

Last-frame predictions



New shot (aligned)

First-frame predictions



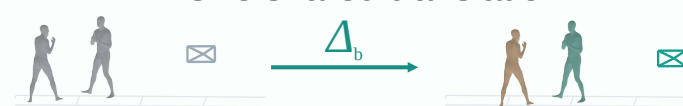
Pelvis position · Torso orientation
Root-centered joint configuration

Matched pairs

Persistent IDs

Person-guided translation

One shared translation



Camera + all humans · relative geometry preserved

Streaming output

World-frame output

Schematic illustration



World coordinates
Cameras · Humans · Identities

Past outputs remain unchanged